

Package ‘ipcr’

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Type Package

Title Individual Parameter Contribution Regression

Version 0.3.3

Description Provides methods for explaining and predicting individual and group differences in model parameters using individual parameter contribution regression (IPCR). The package is primarily designed for structural equation models fitted with the lavaan and OpenMx packages, but also supports lme4, as well as the lm and glm functions from the stats package.

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lmtest,
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OpenMx,
sandwich

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get_ipcs

*Calculate Individual Parameter Contributions***Description**

This functions calculates the individual parameter contributions (IPCs) of a fitted model.

Usage

```
get_ipcs(x, analytic = FALSE, ...)
```

Arguments

x	a fitted model object.
analytic	Logical. If FALSE (default), functions of lavaan , OpenMx , or sandwich will be used to compute scores. If TRUE, custom functions will be used. This is only relevant for models fitted with OpenMx where the computation of the scores can take time. Supports MxRAMModel without algebras.

Details

get_ipcs is a convenience function for obtaining IPCs. The more powerful [ipcr](#) function also provides IPCs and can be used to predict differences in model parameters by regressing the IPCs on covariates.

Value

A data.frame containing the of a fitted model. The dimensions of this data.frame are $n * k$, where n denotes the number of observations and k the number of parameters. The columns should be named as in [coef](#).

References

Arnold, M., Oberski, D. L., Brandmaier, A. M., & Voelkle, M. C. (2019). Identifying heterogeneity in dynamic panel models with individual parameter contribution regression. *Structural Equation Modeling*, 27, 613-628. doi: [10.1080/10705511.2019.1667240](https://doi.org/10.1080/10705511.2019.1667240)

See Also

[ipcr](#)

get_scores	<i>Extract Scores</i>
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Description

This functions extracts the scores of a fitted model. Arguments are passed to the generic function [estfun](#).

Usage

```
get_scores(x, analytic = FALSE, ...)
```

Arguments

x	a fitted model object.
analytic	Logical. If FALSE (default), functions of lavaan , OpenMx , or sandwich will be used to compute scores. If TRUE, custom functions will be used. This is only relevant for models fitted with OpenMx where the computation of the scores can take time. Supports MxRAMModel without algebras.
...	arguments passed to methods.

Value

A data.frame containing the empirical estimating functions. Typically, this should be an $n * k$ matrix corresponding to n observations and k parameters. The columns should be named as in [coef](#) or [terms](#), respectively.

The estimating function (or score function) for a model is the derivative of the objective function with respect to the parameter vector. The empirical estimating functions is the evaluation of the estimating function at the observed data (n observations) and the estimated parameters (of dimension k).

References

Zeileis, A. (2006). Object-oriented computation of sandwich estimators. *Journal of Statistical Software*, 16(9), 1-16. doi: [10.18637/jss.v016.i09](https://doi.org/10.18637/jss.v016.i09)

Zeileis, A., Köll, S., Graham, N. (2020). Various versatile variances: An object- oriented implementation of clustered covariances in R. *Journal of Statistical Software*, 95(1), 1-36. doi [10.18637/jss.v095.i01](https://doi.org/10.18637/jss.v095.i01)

See Also

[estfun](#)

identify_definition_variables

Identifies definition variables in OpenMx models

Description

This is an internal function that identifies which variables in the data set stored in an **OpenMx** model are definition variables.

Usage

```
identify_definition_variables(x)
```

Arguments

x An OpenMx model.

ipcr

Individual Parameter Contribution Regression

Description

Performs individual parameter contribution regression (IPCR), a method for modeling parameter heterogeneity by regressing parameter estimates on predictors. `ipcr` is designed for structural equation models fitted with **lavaan** and **OpenMx**, but it also supports `lm`, `glm`, and `lmer` models.

Usage

```
ipcr(fit, predictors, analytic = FALSE)
```

Arguments

fit	A fitted model object. Supported models include those estimated using lavaan , OpenMx , <code>lm</code> , <code>glm</code> , and <code>lmer</code> .
predictors	A vector, matrix, or <code>data.frame</code> containing one or more predictors used to predict variation in model parameters. Interaction and polynomial terms can be included as new variables, which may require centering. Ensure categorical variables are properly coded as factors or dummy variables.
analytic	Logical. If <code>FALSE</code> (default), functions of lavaan , OpenMx , or sandwich will be used to compute scores. If <code>TRUE</code> , custom functions will be used. This is only relevant for models fitted with OpenMx where the computation of the scores can take time. Supports <code>MxRAMModel</code> without algebras.

Details

Individual parameter contributions (IPCs) provide rough approximations of individual-specific parameter values. The IPCs of individual i is defined as:

$$IPC_i = \theta + A(\theta)^{-1} S(\theta, y_i),$$

where θ represents the estimated model parameters, $S(\theta, y_i)$ is the estimating function (e.g., the first derivative of the log-likelihood), and $A(\theta)$ is the expected negative derivative of the estimating function (often called the Hessian matrix). By regressing IPCs on predictors, the sources of parameter heterogeneity can be examined.

IPCs are known to be slightly biased. This bias can be corrected using iterated IPC regression, which iteratively recalculates the IPCs. The function `ipcr_it` implements iterated IPC regression.

To identify the most important predictors of heterogeneity, regularization can be applied by passing the output of `ipcr` or `ipcr_it` to the `ipcr_reg` function.

Value

An object of class "ipcr", which is a list containing the following elements:

<code>info</code>	A list with metadata about the <code>ipcr</code> function call.
<code>IPCs</code>	A <code>data.frame</code> containing individual parameter contributions.
<code>mlm</code>	An object of class <code>lm</code> (for a single parameter) or <code>mlm</code> (for multiple parameters), representing the regression model.
<code>mancova</code>	An object of class <code>manova</code> (if the model contains multiple parameters) or <code>NULL</code> (if the model contains only a single parameter).
<code>output</code>	A list of formatted output tables, which can be examined using <code>print</code> and <code>summary</code> .

The function `summary` provides an overview of the IPC regression models. Use `summary.ipcr` to extract specific information from the model results. The `print` function displays the arguments specified in the `ipcr` call. The `plot` function visualizes the correlations between IPCs and predictors as a heatmap.

Several generic functions can be applied to `ipcr` objects to extract information from the `mlm` object, including: `coef`, `confint`, `effects`, `fitted`, `nobs`, `predict`, `residuals`, `sigma`, and `vcov`.

Heteroskedasticity-robust IPC regression can be performed using `coeftest` and `coefci` from the **lmtest** package.

References

- Arnold, M., Oberski, D. L., Brandmaier, A. M., & Voelkle, M. C. (2020). Identifying heterogeneity in dynamic panel models with individual parameter contribution regression. *Structural Equation Modeling*, 27, 613-628. doi: [10.1080/10705511.2019.1667240](https://doi.org/10.1080/10705511.2019.1667240)
- Arnold, M., Brandmaier, A. M., & Voelkle, M. C. (2021). Predicting differences in model parameters with individual parameter contribution regression using the R package `ipcr`. *Psych*, 3, 360-385. doi: [10.3390/psych3030027](https://doi.org/10.3390/psych3030027)

See Also

[ipcr_it](#), [ipcr_reg](#)

Examples

```
# Load the Holzinger and Swineford (1939) dataset from the lavaan package
HS_data <- lavaan::HolzingerSwineford1939

# Remove observation 301 because one of its predictor values is missing
HS_data <- HS_data[stats::complete.cases(HS_data), ]

# Define a confirmatory factor analysis (CFA) model using lavaan syntax
# This model specifies three latent factors: visual, textual, and speed
m <- 'visual =~ x1 + x2 + x3
      textual =~ x4 + x5 + x6
      speed  =~ x7 + x8 + x9'

# Fit the CFA model
fit <- lavaan::cfa(model = m, data = HS_data)

# Preprocess the predictor variables

# Combine the year part (ageyr) and the month part (agemo) of age into a single
# variable that measures age in years
HS_data$age <- HS_data$ageyr + HS_data$agemo / 12

# Convert sex and grade into dummy variables (coded as 0 and 1)
HS_data$sex <- HS_data$sex - 1
HS_data$grade <- HS_data$grade - 7

# Create a new data.frame with the predictor variables
predictors <- HS_data[, c("sex", "age", "school", "grade")]

# Perform Individual Parameter Contribution Regression (IPCR)
res <- ipcr(fit = fit, predictors = predictors)

# Plot a heatmap showing correlations between parameters and predictors
plot(res)

# Display a summary of the IPC regression results
summary(res)
```

ipcr_it

Iterated Individual Parameter Contribution Regression

Description

Performs iterated individual parameter contribution regression.

Usage

```
ipcr_it(
  fit,
  predictors,
  analytic,
  conv = 0.01,
  learning_rate = c(seq(from = 1, to = 0.1, by = -0.1), 0.05, 0.01, 0.005, 0.001),
```

```

    max_it = 350,
    iteration_info = FALSE
  )

```

Arguments

fit	A fitted model object. Supported models include those estimated using lavaan and OpenMx .
predictors	A vector, matrix, or data.frame containing one or more predictors used to predict variation in model parameters. Interaction and polynomial terms can be included as new variables, which may require centering. Ensure categorical variables are properly coded as factors or dummy variables.
analytic	Logical. If FALSE (default), functions of lavaan , OpenMx , or sandwich will be used to compute scores. If TRUE, custom functions will be used. This is only relevant for models fitted with OpenMx where the computation of the scores can take time. Supports MxRAMModel without algebras.
conv	an integer used as a stopping criterion for iterated IPC regression. The criterion is the largest difference in any parameter estimate between iterations.
learning_rate	stepsize used to calculate the updated IPCs. It starts with the largest value and uses smaller values until the algorithm converged or no other values are available.

ipcr_reg	<i>Add together two numbers</i>
----------	---------------------------------

Description

Add together two numbers

Usage

```

ipcr_reg(
  fit,
  covariates = NULL,
  iterate = FALSE,
  iteration_info = FALSE,
  conv = 1e-04,
  max_it = 50,
  regularization = FALSE,
  s = "lambda.min",
  alpha = 1,
  weights = NULL,
  nlambdas = 100,
  standardize = TRUE,
  nfolds = 10,
  linear_MxModel = TRUE
)

```

Arguments

x	A number.
y	A number.

Value

A number.

plot.ipcr

Plot Correlations Between Predictors and IPCs

Description

Generates a heatmap visualizing the correlations between predictors and individual parameter contributions (IPCs).

Usage

```
## S3 method for class 'ipcr'
plot(x, print_corr = FALSE, ...)
```

Arguments

x	An object of class "ipcr".
print_corr	Logical; if TRUE, correlation coefficients are displayed on the heatmap. Default is FALSE.
...	Additional arguments (currently unused).

Details

This function is a wrapper for [ggplot](#). Currently, arguments passed to ggplot cannot be modified.

plot_differences

Plot Estimated Conditional Differences in Model Parameters

Description

Generates plots of estimated model parameters as a function of one or more predictors.

Usage

```
plot_differences(
  x,
  parameter = NULL,
  predictor = NULL,
  confidence_level = 0.95
)
```


Arguments

x	An object of class "ipcr".
parameter	A character string specifying the name of a model parameter as displayed by print. By default, all model parameters are plotted.
predictor	A character string specifying the name of a predictor. By default, the effects of all predictors are plotted.
confidence_level	A numeric value indicating the confidence level for the confidence intervals. The default is 0.95, corresponding to a 95 percent confidence interval.

Details

The function plots regression lines for continuous predictors and group means for dummy variables. Confidence intervals are included based on the model parameter estimates and their corresponding variances and covariances. The plots visualize the estimated parameter values as a function of a predictor, assuming all other predictors are set to zero. Mean-centering predictors may improve interpretability. This function is a wrapper for [ggplot](#).

See Also

[plot.ipcr](#)

Examples

```
# Generate data
## predictors
z1 <- rep(0:1, each = 50)
z2 <- rnorm(n = 100)
predictors <- data.frame(z1 = z1, z2 = z2)
## Model data
x <- rnorm(n = 100)
y <- 0.5 + 0.75*z1 + rnorm(n = 100, sd = sqrt(0.75))
d <- data.frame(x = x, y = y)

# Fit a linear regression
m <- lm(y ~ x, data = d)

# Investigate model with IPC regression
res <- ipcr(m, predictors = predictors)

# Plotting parameter values as a function of the predictors
## All plots
plot_differences(res)
## Plot the values of the regression slope x as a function of the predictor z1
plot_differences(res, parameter = "x", predictor = "z1")
```

`summary.ipcr`*Individual Parameter Contribution Regression Summary*

Description

Summary method for objects of class "ipcr".

Usage

```
## S3 method for class 'ipcr'  
summary(object, what = "everything", digits = 3, ...)
```

Arguments

<code>object</code>	An object of class "ipcr".
<code>what</code>	A character string specifying what to display. Options are: • "everything" (default): Displays all summary components. • "marginal": Shows only marginal linear effects. • "ipcr": Shows only individual parameter contribution regression coefficients.
<code>digits</code>	An integer indicating the number of decimal places to use in the output. Default is 3.
<code>...</code>	Additional arguments passed to or from other methods.

Value

Invisibly returns NULL.

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